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SUPERSEDING
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MILITARY SPECIFICATION

TESTER, CYLINDER COMPRESSION (AUTOMOTIVE MAINTENANCE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an automotive maintenance cylinder compression tester used to determine internal combustion engine cylinder compression not exceeding 250 pound-force per square inch gage (psig). The cylinder compression tester is of the universal type with threaded adapters and rubber cones.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898	-	Cushioning Material, Packaging (Cellulosic, Water Absorbant)
PPP-B-566	-	Box, Folding, Paperboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-60	-	Tape: Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-R-3065	-	Rubber, Fabricated Products
MIL-C-5501	-	Caps and Plugs, Protective, Dust and Moisture Seal General Specification for.
MIL-R-6855	-	Rubber, Synthetic Sheets, Strips, Molded or Extruded Shapes General Specification for
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-101	-	Test Procedures for Packaging Materials
FED-STD-151	-	Metal; Test Methods

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications are available from the Defense Printing Service Detachment Office, Building 4D NPM-DoDSSP, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the document cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E380	-	Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents are also available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Design. The cylinder compression tester shall be a portable instrument consisting of a dial gage, valve assembly, adapters, connectors, and a carrying case. The dead air space in the components and assemblies shall be minimized to insure correct reading after four compression strokes. The tester shall indicate compression psig directly within the engine cylinders.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the cylinder compression tester. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380.

3.2.2 Maintainability. Design and configuration of the cylinder compression tester and accessories shall assure that maintenance of operational capability will be reduced to the simplest form or procedures and will require a minimum of common maintenance equipment. Instructions for disassembly, repair, and assembly shall be clear, concise, and definitive in application.

3.2.3 Interchangeability. All parts of the cylinder compression tester shall be manufactured to definitive standards and tolerances that will provide for interchangeability of respective parts between cylinder compression testers of the same model. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the cylinder compression tester, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. The cylinder compression tester may contain reclaimed materials provided the materials will not jeopardize the cylinder compression tester's intended use or performance. The reclaimed material shall be reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the cylinder compression tester. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. All parts and materials used in fabricating the cylinder compression tester shall be free of defects. In no event shall patching processes be used to reclaim defective parts. External surfaces shall be smooth and free of sharp edges. The construction of the cylinder compression tester shall be rugged and sturdy and commensurate with the current practice within the professional commercial automotive test equipment industry.

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3.4.1 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.2 Corrosion resistance. All exposed metal and working surfaces, except those made of corrosion resistant materials, shall be coated or covered with a protective finish that provides protection against corrosion.

3.5 Performance and product characteristics.

3.5.1 Pressure. All torqued or coupled connections shall withstand the maximum air pressure for which the scale is graduated. There shall be no leakage when torqued to 3 ± 0.5 pound feet or coupled with a quick-connect coupler.

3.5.2 Accuracy. The tester shall comply with the accuracy requirements specified in Table I.

TABLE I. Allowable error in pressure indication.

Conditions	Center half of indicator scale (Max. error)	Upper & lower quarters of indicator scale (Max. error)
(a) Initial accuracy - testers stabilized at +70 DEG. to +90 DEG. Fahrenheit (F)	4 psig	6 psig
(b) High temperature accuracy - testers stabilized at +110 DEG. to +115 DEG. F	5 psig	12 psig
(c) Low temperature accuracy - testers stabilized at +30 DEG. to +35 DEG. F	5 psig	12 psig
(d) Accuracy after all tests - testers stabilized at +70 DEG. to +90 DEG. F	5 psig	7 psig

3.5.3 Vibration. The cylinder compression tester and all components enclosed in the carrying case shall show no damage and shall meet the requirement of Table I (condition d) after being subjected to a vibration of simple harmonic motion at a frequency range of 5 to 55 Hertz (Hz), with a maximum amplitude of 0.030 inch (0.060 inch maximum total excursion), through each of the three major axes for a period of 30 minutes in each axis.

3.5.4 Shock. The cylinder compression tester and all components enclosed in the carrying case shall show no damage and shall meet the requirement of Table I (condition d) after being dropped three times, once through each of the three major axes, on a concrete floor from a height of no less than eight inches.

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3.5.5 Operating temperatures. The cylinder compression tester shall meet the requirements of Table I (condition b) while stabilized at +110 degrees to 115 degrees F, and the requirements of Table I (condition c) while stabilized at +30 degrees to +35 degrees F.

3.5.6 Endurance. The tester shall exhibit no functional failure and shall meet the requirements of Table I (condition d) after being subjected to 1,000 operating cycles, at a rate of ten cycles per minute (cpm), at each of the following pressures: 75, 125, 175, and 225 psig.

3.6 Component details.

3.6.1 Gage. The cylinder compression tester shall employ a dial indicating, bourdon tube type pressure gage.

3.6.1.1 Operating mechanism. The bourdon tube shall be of phosphor bronze, beryllium copper, or stainless steel. The tube shall be secured into the socket by solder or by being threaded and soldered. The end opposite the socket end shall be sealed by a soldered or threaded and soldered plug.

3.6.1.2 Stem. The tube and other operating mechanism of the gage shall be supported by the stem which shall be securely connected to the housing in such a manner that working or twisting the housing shall not cause movement of the pointer. An integral wrench grip, extending at least 0.25 inch outside the housing, shall be provided.

3.6.1.3 Dial. The dial shall be not less than 2.5 inches in diameter with a scale arc of not less than 270 degrees. Graduations and figures shall be black on a dull white background or white on a black background. Black dial mark graduations shall not exceed .04 inches in width. The dial scale shall have a minimum scale range of 0 to 250 pound-force per square inch gage (psig). The dial scale shall be graduated in increments of not more than five psig, with numerical markings at least every 25 psig interval.

3.6.1.4 Pointer. The pointer shall be plain and shall be black or red for a dial with white background and white for a dial with a black background. The width of the pointer tip shall not exceed .04 inches.

3.6.1.5 Housing. The gage housing shall be of all metal construction and shall have a threaded or screw retained bezel to retain the crystal. The bezel shall be of the same material as the body of the housing. Openings shall be provided in the housing to relieve the pressure in the event of bourdon tube failure.

3.6.1.6 Crystal. The crystal shall be not less than 0.052 inch thick and shall be of either transparent plastic material, unaffected by oil and gasoline, or of shatterproof glass.

3.6.2 Valve assembly. The valve assembly shall be designed to screw onto, or into, the gage stem to form an air pressure tight connection. An integral wrench grip, at least .375 inch wide, shall be provided. The opposite end of the valve assembly shall be threaded to form an air pressure tight joints with each adapter or connector. Metal parts of the valve assembly shall be quarter or half hard temper brass, steel, or aluminum anodized in accordance with MIL-A-8625, type II. Steel parts shall be coated with corrosion-resistant material and capable of withstanding a 48 hour salt spray test in accordance with FED-STD-151, method 811. The valve assembly shall contain a check valve to retain the pressure admitted to the gage until

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released by the operator. A relief valve shall be provided either in the valve assembly or inserted in that part of the gage which connects to the valve assembly to release retained pressure and return the gage pointer to zero. The relief valve shall be manually operated, externally accessible, requiring no disassembly of the tester and no tools to operate, and capable of being operated without removing the tester from the engine spark plug hole. One valve may be used to perform both the check valve function and the relief valve function. The valve, or valves, and gaskets shall be made of oil-resistant material equal to or exceeding the requirements of MIL-R-3065, type S, class SA, or MIL-R-6855, class 1, grade 80. The valve shall operate satisfactorily after being immersed in SAE No. 10 motor oil for a period of 24 hours. The loss of air pressure from the valve assembly during a five minute period, after an initial pressure of 150 psig, shall be not more than five psig.

3.6.3 Universal adapter set. The universal adapter set shall consist of three metal adapters with rubber cones, four metal threaded adapters, and one flexible adapter extension.

3.6.3.1 Metal adapters - rubber cones. The metal adapters shall be such that air-tight connections can be made between adapters and the valve assembly with gage. One metal adapter shall be 3 inches ± .25 inch in length, one shall be 6.25 inches ± .25 inch, and one shall be 6 inches ± .25 inch with a "V" bend of 135 degrees located 3 inches, or halfway, from the end. One end of the metal adapters shall incorporate the rubber cones, the other end shall receive the valve assembly with gage or the flexible extension and shall have an integral wrench grip at least .375 inch wide. The adapters shall be quarter or half hard temper brass, steel, or aluminum anodized in accordance with MIL-A-8625, type II. Steel parts shall be coated with corrosion-resistant material and capable of withstanding a 48 hour salt spray test in accordance with FED-STD-151, method 811. The rubber cones shall fit 10mm, 14mm, 18mm, and 7/8 inch spark plug ports. Cone material shall conform to the requirements of MIL-R-6855, class 1, grade 70. The rubber cones and the metal adapters, including the threaded gage fittings and tubes, shall be capable of withstanding a force of 45 pounds when applied against a threaded hole simulating a standard spark plug port. When removed from the threaded hole, the cones and adapters shall show no damage, splitting rubber separation, bent tubes, or damaged threads.

3.6.3.2 Metal adapters - threaded. Threaded adapters shall be furnished to fit 10mm, 14mm, 18mm, and 7/8 inch standard spark plug ports. A 13/16 inch hex wrench grip, or hex grip of the same size normally furnished on the corresponding spark plugs, at least .75 inch wide shall be provided. The overall length of each adapter shall be 2.375 inches ± .125 inch. The size of the spark plug ports shall be legibly marked on each adapter in accordance with MIL-STD-130. The spark plug port end threaded length shall be .375 inch ± .0625 inch and the other end shall be threaded to receive the valve assembly and the flexible extension. The adapters shall be quarter or half hard temper brass or steel. Steel parts shall be coated with corrosion-resistant material and capable of withstanding a 48 hour salt spray test in accordance with FED-STD-151, method 811.

3.6.3.3 Flexible adapter extension. The overall length of the flexible extension shall not be less than eight inches. The fittings at each end shall provide a .75 inch wrench grip at least .25 inch wide. One end shall be threaded to receive the adapter and the other end to receive the valve assembly connection. Threaded fittings shall be quarter or half hard temper brass, steel, or aluminum anodized in accordance with MIL-A-8625, type II. Steel parts shall be coated with corrosion-resistant material and capable of withstanding a 48 hour salt spray test in accordance with FED-STD-151, method 811. The flexible extensions shall be fabricated of flexible oil-resistant rubber hose. The hose and fittings shall be capable of withstanding a

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hydrostatic proof pressure of 600 psig for a period of ten minutes, after being submerged in SAE No. 10 motor oil for a period of 24 hours, without evidence of bulging, separation of fittings, splitting, bursting, deterioration, or deformation.

3.6.4 Carrying case. A carrying case, to store and transport the cylinder compression tester and all of its component parts, shall be fabricated from either sheet steel or high density polyethylene. The overall size of the case shall not exceed 150 cubic inches.

3.6.4.1 Steel carrying case. The steel case shall be manufactured from no less than 21 gage (0.033 inch thick) steel sheet, with a hinged lid and a positive type fastening latch. The case shall have means to secure the gage and each adapter within the case.

3.6.4.2 High density polyethylene case. The polyethylene case shall be constructed of high density polyethylene (which is a polyethylene with a specific gravity of 0.941 to 9.965), which is blow molded or injection molded to the desired shape, with a contoured interior to retain each component. The case shall have a carrying handle and positive closure clasps. The case shall have a lid that opens on a mechanical style hinge consisting of a male plastic pin fitting into a female dent, forming a swivel socket configuration. Living hinges, hinges consisting of a flexible flap of polyethylene extending from the lid to base, are not acceptable. No metal parts shall be used in the construction of the case including the handle, clasps, hinge, feet, marking or fasteners.

3.7 Cranking switch. A cranking switch shall be provided to allow engine cranking by one operator while testing compression. The cranking switch shall be of sufficient size to handle 200 amperes for 30 seconds while operating at either 12 or 24 volts (DC). The cranking switch shall be provided with two rubber covered, twin-flex, No. 10 AWG leads at least six feet in length. The leads shall be terminated on one end by the switch and on the other end with one red insulated and one black insulated alligator clamp for connection to the starter solenoid and battery.

3.8 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with the current professional commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.9 Product identification. The cylinder compression tester shall be marked on the lid of the carrying case for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.10 Workmanship. Standards of workmanship shall assure that the cylinder compression tester have the stability, strength, safety, and efficient operating characteristics found in current professional commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government

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reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production cylinder compression testers being offered for acceptance under the contract. The inspection shall consist of (a) through (d) as follows. Failure of any cylinder compression tester to pass these examinations, inspections, or tests shall be cause for rejection of the cylinder compression tester.

(a) Product examination (see 4.5).
(b) Accuracy test (see 4.6.1).
(c) Pressure retention test (see 4.6.2).
(d) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or Test	Lot or Batch Size	Sample Size
Product examination (see 4.5)	1-150	13
Accuracy test (see 4.6.1)	1-150	13
Pressure retention test (see 4.6.2)	1-150	13
Packaging inspection (see 4.7)	1-150	5
Rough handling test (see 4.8)	1/	1/

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100 percent inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

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1/ When first article test is specified, the rough handling test shall be performed on the first article only. If no first article is required, the rough handling test shall be performed on one unit pack selected randomly from the first lot of the total quantity of cylinder compression testers.

4.5 Product examination. Visually, dimensionally, and manually examine each cylinder compression tester to determine conformance with the requirements of 3.2 through 3.4.2 and 3.6 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the tester and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definitive standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the cylinder cylinder compression tester (see 3.2.3), shall be cause for rejection.

4.6 Tests.

4.6.1 Accuracy. The cylinder compression tester shall be connected with the adapter assemblies and shall be assembled. The cylinder compression tester components shall be tightened to each other with a torque of 3 ± 1/2 pound-feet (see 3.5.1). Accuracy shall be tested at all graduations with numerical markings, throughout the entire range of scale, by comparison with a standard air pressure measuring device having an overall accuracy within one percent. Accuracy shall be within the applicable limits specified in Table I (condition a).

4.6.1.1 Accuracy after all tests. The cylinder compression tester shall be installed on a cylinder of a multi-cylinder engine. Three consecutive pressure readings between 75 and 250 psig shall be taken, after four compression strokes, with the throttle open and all spark plugs removed. The pressure obtained with the gage shall be compared with the pressure indications of a gage of known accuracy. The accuracy shall be within the limits specified in Table I (condition d).

4.6.2 Pressure retention. An air pressure of 150 psig shall be applied for a minimum of six seconds and then, after a five minute interval, the gage indication shall be examined for air pressure loss. Pressure loss shall not exceed five psig (see 3.6.2).

4.6.3 Vibration. The cylinder compression cylinder compression tester and all components enclosed in the carrying case shall be rigidly mounted on a vibration machine and subjected to vibration (simple harmonic motion) with an amplitude of 0.030 inch (0.060 inch total excursion) through a frequency range of 5 to 55 Hz, for a period of 30 minutes each, in the direction of each of the three major axes. The frequency shall be continually varied with the entire range being accomplished once every five minutes. At the conclusion of the test, the cylinder compression tester shall be examined for evidence of damage and tested in accordance with 4.6.1, table I (condition d) to determine compliance with 3.5.3.

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4.6.4 Shock. The cylinder compression tester and all components enclosed in the carrying case shall be dropped three times, once through each of the three major axes, to a concrete floor from a height of eight inches. On conclusion of the test, the cylinder compression tester shall be examined for evidence of damage and tested in accordance with 4.6.1, table I (condition d) to determine compliance with 3.5.4.

4.6.5 Salt spray. The valve assembly (if fabricated of steel) and the adapter set (if fabricated of steel) shall be subjected to the salt spray test specified in FED-STD-151, method 811, for 48 hours. At the end of this test period, the valve assembly and the adapter set shall be washed free of salt particles, air dried to remove surface moisture, and examined for evidence of damage (see 3.6.2, 3.6.3.1, and 3.6.3.3). The cylinder compression tester shall then be tested for accuracy in accordance with 4.6.1, table I (condition d).

4.6.6 Valve. The valve or valves shall be completely immersed in SAE No. 10 motor oil for a period of 24 hours. At the end of the immersion time, the valve or valves shall be removed from the oil, cleaned, and reassembled to the cylinder compression tester. The cylinder compression tester shall be tested in accordance with 4.6.2 to determine compliance with 3.6.2.

4.6.7 Flexible extension. The flexible extension shall be completely immersed in SAE No. 10 motor oil for a period of 24 hours. At the conclusion of the immersion period, the flexible extension shall be cleaned and subjected to a hydrostatic pressure of 600 psig. During the pressure test, the flexible extension shall be observed, and on conclusion of the test the flexible extension shall be examined to determine compliance with 3.6.3.3.

4.6.8 Operating temperature.

4.6.8.1 High. The complete cylinder compression tester, after being in an ambient air temperature of +110 degrees to +115 degrees F until stabilized and while at that temperature, shall be tested in accordance with 4.6.1, table I (condition b) to determine compliance with 3.5.5.

4.6.8.2 Low. The complete cylinder compression tester, after being in an ambient air temperature of +30 degrees to +35 degrees F until stabilized and while at that temperature, shall be tested in accordance with 4.6.1, table I (condition c) to determine compliance with 3.5.5.

4.6.9 Endurance. At the conclusion of the oil-resistant valve test specified in 4.6.6, the cylinder compression tester shall be assembled as specified in 4.6.1 and air pressure shall be applied for approximately one second on and five seconds off, with the air pressure relieved by operating the cylinder compression tester relief valve once each cycle, and prior to application of pressure for the next test cycle. The cylinder compression tester shall be subjected to 1,000 cycles at a rate of 10 cpm at each of the following pressures: 75, 125, 175, and 225 psig. The cylinder cylinder compression tester shall then be inspected for evidence of damage and tested in accordance with 4.6.1, table I (condition d) to determine compliance with 3.5.6. After completion of this test, the cylinder compression tester shall be subjected to the test in 4.6.10.

4.6.10 Force. Apply a 45 pound force to cones sized to fit against a 10mm, 14mm, 18mm and 7/8 inch threaded hole simulating a spark plug port. Apply a 45 pound force for 15 continuous seconds. The cones and adapters shall show no damage, splitting rubber separation, bent tubes, or damaged threads after removal of the cone.

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4.6.11 Cranking switch. Apply 12 Volts (DC) to the cranking switch. The cranking switch shall operate at 200 amperes for 30 seconds while operating a starter at either 12 or 24 volts (DC). The cranking switch shall show no damage.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the preservation, packaging, packing and marking requirements specified herein.

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4.8 Rough handling. After the quality assurance inspection and tests have been successfully completed, one item, packaged in accordance with the packaging requirements specified herein, shall be selected at random and subjected to the following rough handling tests specified in FED-STD-101. The apparatus used shall be as described in FED-STD-101. A dummy load shall not be used. All tests are to be conducted under ambient conditions. Environmental temperature and humidity tests are not required. After the test has been completed, the container shall be inspected. Defects in accordance with 4.8.1 and FED-STD-101 shall be cause for rejection. The unit shall then be unpacked. If there is evidence of possible damage to the cylinder compression tester which cannot be visually determined, the tester shall again be subjected to the accuracy test (see 4.6.1) to the requirement of table I, condition (d), and pressure retention test (see 4.6.2). After all requirements have been met, the unit shall be inspected and repackaged (see 4.7) before submission for government acceptance. After testing, any damage to the packaging shall be replaced prior to acceptance of the lot.

4.8.1 Mechanical handling. The mechanical handling test shall be conducted using the lifting and transportation by forklift truck procedure of Method 5011.1 of Fed. Test Method Std. No. 101. Criteria for passing or failing the mechanical handling test shall be as follows:

"When the container or package is subjected to the mechanical handling test, the contents (except a dummy load) shall show no functional or physical damage, and the container and packing shall show no functional damage. Damage to the exterior shipping container which is the result of improper interior packaging, blocking, or bracing shall be cause for rejection. Structural failure of the exterior shipping container which would result in spilling of the contents or failure of the container in subsequent handling is cause for rejection. There shall be no evidence of a substantial amount of shifting of the contents within the exterior shipping container that would create conditions likely to cause damage during shipment, storage, and reshipment of the container. Minor container damage such as chipping of wood members, negligible dents, paint chipping, is not cause for rejection."

5 PACKAGING

5.1 First article. When specified (see 6.2), the supplier shall furnish one complete pack for first article packaging inspection. The first article may be either a preproduction pack or an initial production pack which conforms to the requirements stated herein. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements stated herein.

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5.2 Preservation/packaging. Preservation/packaging shall be Level A, C or Commercial/industrial, as specified (see 6.2). Unless otherwise specified (see 6.2), military level C shall be provided.

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The cylinder compression tester shall be cleaned and dried in accordance with process C-1 and procedures D-1 or D-4 of MIL-P-116, respectively.

5.2.1.2 Unit packaging. Each cylinder compression tester shall be packaged in accordance with method IC-2 of MIL-P-116. Exposed threaded surfaces shall be protected against physical damage with tape conforming to PPP-T-60, type optional, or plastic caps or plugs conforming to MIL-C-5501. The gage crystal shall be protected against damage by wrapping with cushioning material conforming to A-A-1898. The cylinder compression tester components shall be placed in their respective compartments or retaining brackets in the carrying case. Cushioning material shall be provided to immobilize the components within the carrying case. A box conforming to PPP-B-566, variety 1, style and class optional, or PPP-B-636, class domestic, shall be used as the unit container. Technical publications, when applicable, shall be placed in a kraft paper or plastic envelope and placed inside the method IC-2 package.

5.2.2 Level C. Level C preservation/packaging shall be in accordance with the requirements of MIL-STD-1190.

5.2.3 Commercial/industrial. Commercial/industrial packing shall be in accordance with the requirements of ASTM D3951.

5.3 Packing. Packing shall be Level A, B, C or Commercial/industrial, as specified (see 6.2). If no level is specified, military level C shall be provided.

5.3.1 Level A. Testers shall be packed in a shipping container conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. Anchoring, blocking, bracing and cushioning shall be in accordance with MIL-STD-1186. The gross weight and size of the exterior shipping container shall be governed by the limitations of the applicable container specification.

5.3.2 Level B. Testers shall be packed as specified in 5.3.1 except the shipping containers shall conform to PPP-B-601, domestic type, or PPP-B-621, class 1.

5.3.3 Level C. Level C packing shall be in accordance with the requirements of MIL-STD-1190.

5.3.4 Commercial/industrial. Commercial/industrial packing shall be in accordance with the requirements of ASTM D 3951.

5.4 Marking. All markings, whether military or commercial, for interior and exterior containers shall be in accordance with the requirements of MIL-STD-129.w

5.5 Rough handling. The completed unit pack, shipping container(s) and all contents shall be capable of withstanding the shipping and handling conditions expected from fork lift handling and truck transport. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling test(s) specified herein.

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6. NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cylinder compression tester is used to measure the compression of spark ignition, internal combustion engines. The cylinder compression tester valve assembly maintains the engine pressure as required, then is used to release the entrapped pressure. The flexible and the universal adapter sets provide adapters to adapt the cylinder compression tester to the spark plug ports of all spark ignition, internal combustion engines with 10mm, 14mm short and long with flat seats and tapered seats, 18mm with flat seats and tapered seats, and 7/8 inch ports.

6.2 Acquisition requirements . Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1 and 5.1).
(c) Painting and finishing if different (see 3.8).
(d) Product identification if different (see 3.9).
(e) First article inspection if different (see 4.3).
(f) Level of preservation, packing, and marking if different (see 5.2,
(g) Level of preservation, packing, and marking if different (see 5.2,

6.3 First article. When a first article inspection is required, the cylinder compression tester should be a first article sample, a first production item, a sample selected from the first run of production items or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of a specific number of units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4.1 Repair part manuals. Repair part identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Commercial and Government Entity Code (CAGE), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

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6.5 Subject term (key word) listing.

Compression
Engine Cylinder
Tester

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Air Force - 99

Review activities:

Army - ME
Air Force - 82
DLA - CS

Preparing activity:

Army - AL

Project No. 4910-0721

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-13011H	930514

3. DOCUMENT TITLE

TESTER, CYLINDER COMPRESSION (AUTOMOTIVE MAINTENANCE)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
DALE HOLLAND	309-782-8146	793-8146
c. ADDRESS (Include Zip Code) COMMANDER US ARMY ARMAMENT RESEARCH, DEV & ENGR CTR ATTN: SMCAR-EST-P ROCK ISLAND, IL 61299-7300	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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